gpeyt1-06 · OC clay settlement

GEO PE · Long Calculation Problems I — Topic 2: Soil Mechanics, Lab Testing & Analysis

Problem

A warehouse slab will add a uniform Δσ = 1,200 psf (wide load) at the top of a 16 ft clay layer. Profile: 0–10 ft sand, γ = 115 pcf (water table at 10 ft, top of clay); 10–26 ft clay, γsat = 108 pcf, e0 = 1.05, Cc = 0.42, Cr = 0.06, σ′p = 1,450 psf; below is dense gravel. The clay's cv = 0.35 ft²/day and LL = 61 (not needed here). Use one layer evaluated at mid-depth.

Most nearly, the primary consolidation settlement of the clay is:

A. 10.6 in.
B. 10.0 in.
C. 1.4 in.
D. 8.0 in.

Given equations / data

> γw = 62.4 pcf; OC clay with σ′0 < σ′p < σ′f: S = [Cr H/(1+e0)] log(σ′p/σ′0) + [Cc H/(1+e0)] log(σ′f/σ′p)

Worked solution

1. σ′0 at mid-clay = 10(115) + 8(108 − 62.4) = 1,515 psf
2. σ′f = 1,515 + 1,200 = 2,715 psf > σ′p = 1,450 → crosses σ′p
3. Recompression: 0.06×16/2.05×log(1450/1515) = -0.0089 ft
4. Virgin: 0.42×16/2.05×log(2715/1450) = 0.8928 ft
5. S = 0.8840 ft = 10.6 in.

Final answer

10.6 in.

Why the other values are wrong

- **10.0 in.** — treated the clay as normally consolidated (Cc from σ′0)
- **1.4 in.** — used Cr for the whole stress range
- **8.0 in.** — used total stress at mid-clay

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.2.1–3.2.2 Consolidation (NC and OC soils), pp. 82–84

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=IGS94AWmPWA, https://www.youtube.com/watch?v=NwGztxby-vQ (no creator values copied).

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gpeyt1-07 · Time rate of consolidation

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Problem

An 18 ft layer of soft clay lies between a sand blanket above and a fractured, free-draining sandstone below. A preload is placed instantly. Lab data: cv = 0.28 ft²/day, Cc = 0.38, e0 = 1.12. Ultimate primary settlement under the preload is estimated at 9.6 in. The contractor wants to know how long the preload must stay to reach 90% average consolidation.

Most nearly, the time for U = 90% is:

A. 981 days
B. 57 days
C. 27 days
D. 245 days

Given equations / data

> t = Tv Hd²/cv; Hd = longest drainage distance; Tv(U = 90%) = 0.848, Tv(50%) = 0.197

Worked solution

1. Double drainage (sand top, free-draining rock base) → Hd = H/2 = 9 ft
2. t90 = 0.848 × 9² / 0.28 = 245 days (≈ 0.67 yr)
3. Settlement at that time ≈ 0.9 × 9.6 = 8.6 in. (extra data not needed)

Final answer

245 days

Why the other values are wrong

- **981 days** — used Hd = full thickness (single drainage)
- **57 days** — used Tv for 50%
- **27 days** — forgot to square Hd

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.2.3 Time Rate of Settlement and U–Tv table, p. 85

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gpeyt1-08 · CD triaxial strength

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Problem

A consolidated-drained (CD) triaxial test on a remolded clean sand ($c' = 0$) consolidated to $\sigma'_3 = 20$ psi fails at a deviator stress of 52 psi. Pore pressure was zero throughout; specimen diameter 2.8 in., height 5.6 in.; axial strain at failure 6%. A second specimen of the same sand will be tested at $\sigma'_3 = 35$ psi.

Most nearly, the deviator stress at failure expected for the 35 psi test is:

A. 91 psi
B. 126 psi
C. 99 psi
D. 45 psi

Given equations / data

> $\sin\varphi' = (\sigma'_1 - \sigma'_3)/(\sigma'_1 + \sigma'_3)$; $\sigma'_1 = \sigma'_3 \tan^2(45 + \varphi'/2)$ for $c' = 0$

Worked solution

1. Test 1: $\sigma'_1 = 20 + 52 = 72$ psi
2. $\sin\varphi' = 52/92 = 0.5652 \rightarrow \varphi' = 34.4°$
3. Kp-type ratio $\tan^2(45 + \varphi'/2) = 3.600$
4. Test 2: $\sigma'_1 = 35 \times 3.600 = 126.0$ psi; $\Delta\sigma_f = 126.0 - 35 = 91.0$ psi

Final answer

91 psi

Why the other values are wrong

- **126 psi** — reported σ'_1 instead of deviator stress
- **99 psi** — used $\tan\varphi = \Delta\sigma/\sigma_1$
- **45 psi** — reported the Mohr-circle radius (½ deviator)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.3.2 Shear Strength — Effective Stress (Mohr–Coulomb), p. 87

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gpeyt1-09 · Falling-head k

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Problem

A falling-head permeability test is run on a silty sand specimen 2.5 in. in diameter and 6.0 in. long. The standpipe inside diameter is 0.25 in. The head drops from 40 in. to 24 in. in 26 min. Water temperature 20°C; specimen dry unit weight 104 pcf; Gs = 2.66.

Most nearly, the hydraulic conductivity k is:

A. 5.0e-05 cm/s
B. 2.2e-05 cm/s
C. 2.0e-05 cm/s
D. 5.0e-04 cm/s

Given equations / data

> k = (a L)/(A t) · ln(h1/h2); a = standpipe area, A = specimen area

Worked solution

1. a = π/4 (0.25)² = 0.04909 in.²; A = π/4 (2.5)² = 4.9087 in.²
2. t = 26 × 60 = 1560 s
3. k = (0.04909×6.0)/(4.9087×1560) × ln(40/24) = 1.965e-05 in./s
4. × 2.54 = 4.99e-05 cm/s

Final answer

5.0e-05 cm/s

Why the other values are wrong

- **2.2e-05 cm/s** — used log10 instead of ln
- **2.0e-05 cm/s** — left the result in in./s but labeled cm/s
- **5.0e-04 cm/s** — used diameter ratio d/D instead of area ratio a/A

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.8.5 Permeability Testing, p. 139

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gpeyt1-10 · Δσ′ from drawdown

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Problem

A uniform silty sand deposit is 50 ft thick over rock. Above the water table γ = 110 pcf (moist); below, γsat = 122 pcf. Dewatering for a nearby excavation will lower the water table from 4 ft to 14 ft below grade over several months. Assume the sand above the new water table drains to the moist unit weight and no capillary rise. A settlement-sensitive pipeline rests on piles tipped at 40 ft.

Most nearly, the increase in vertical effective stress at 40 ft is:

A. 624 psf
B. 504 psf
C. 120 psf
D. 1,100 psf

Given equations / data

> σ′ = σ − u; u = γw(z − zw); γw = 62.4 pcf

Worked solution

1. Before: σ′ = 4(110) + 36(122−62.4) = 2,586 psf
2. After: σ′ = 14(110) + 26(122−62.4) = 3,090 psf
3. Δσ′ = 504 psf = 10 ft × (110 − 122 + 62.4)

Final answer

504 psf

Why the other values are wrong

- **624 psf** — took Δσ′ = γw Δzw (ignored the drop in total stress)
- **120 psf** — used only (γsat − γmoist)Δz
- **1,100 psf** — added only the moist weight and forgot buoyancy before

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.3 Effective and Total Stresses, p. 87

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